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**Kanyamanan: A Web-Based Smart Culinary Itinerary and Restaurant Aggregator
System**

A Project Presented to
the Faculty of the Information Technology Department
College of Computing Sciences and Information Technology

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CHAPTER I: THE PROBLEM AND ITS SETTING

1.1 Introduction

In the contemporary global tourism and hospitality sectors, the evolution of experiential travel has positioned culinary tourism as a primary driver of regional economic growth and cultural preservation (World Tourism Organization [UNWTO], 2023). Modern travelers increasingly reject generic tourist itineraries in favor of highly localized, authentic gastronomic experiences (Hall & Gössling, 2022). This structural shift has forced the food and beverage industry to transition from static online listings toward integrated, data-driven digital ecosystems that connect independent culinary merchants into a unified, informative operational grid. Regionally, the rapid expansion of the digital economy across Southeast Asia has highlighted the necessity for localized web frameworks capable of consolidating multi-vendor data, solving real-time logistical constraints, and optimizing traveler movement across broader geographic bounds (Gavalas et al., 2022). Within the Philippine micro-, small-, and medium-enterprise (MSME) sector, there is a critical requirement for collaborative digital platforms that aggregate decentralized food heritage data into an enterprise-grade routing and wellness-tracking infrastructure to enhance market visibility without imposing complex technical onboarding barriers or operational overhead on local business owners (Melnick et al., 2022).

Known for its deeply rooted culinary traditions, the province attracts massive influxes of weekend food tourists, international travelers, and local transit commuters navigating its sprawling municipal networks. However, this widespread provincial demand introduces severe logistical friction for visiting consumers. Tourists attempting multi-town culinary tours frequently face uncoordinated, exhausting travel routes, unpredictable dining queues, and unoptimized travel schedules across Pampanga's 19 municipalities and three component cities. This structural challenge directly aligns with the Personalized Tourist Trip Design Problem (PTTDP), where travel itineraries must dynamically

balance geographic distance with multiple time-dependent stop constraints (Marpaung et al., 2026).

To resolve this, modern travel planning requires real-time geographic solutions. By integrating a live GPS map routing interface with dynamic Estimated Time of Arrival (ETA) tracking, users can actively trace their movement directly on the web application until they reach their destination, mimicking standard, high-usability mapping engines (Marron et al., 2025). Simultaneously, the system enriches the traveler's experience by dynamically displaying and plotting Kapampangan cultural, historical, and ecological attractions located directly along their calculated route, such as the St. James the Apostle Parish in Betis, Guagua, or the San Guillermo Parish Church in Bacolor. This ensures that Pampanga's food map acts not merely as a commercial listing, but as an interactive gateway to the province's holistic cultural landscape (Briola et al., 2025).

Despite the widespread use of generic travel finders, mapping services, and dietary tracking applications, a significant technological gap remains in software designed to manage wide-scale, multi-vendor culinary ecosystems through intelligent data aggregation. Standard directory applications are engineered for standalone search queries and completely lack the spatial mapping logic required to calculate multi-stop, cross-municipality travel itineraries based on shifting venue operational hours, localized crowd density patterns, and individual dietary preferences (Ahmad & Halim, 2023). Furthermore, traditional platforms fail to provide a unified framework capable of handling cross-vendor data synchronization, automated nutritional item breakdown, or vision-based food metric estimation across independent culinary options (Thierry et al., 2021). This technological disconnect forces culinary travelers to rely on manual, error-prone coordination methods, leading to severe traffic congestion around popular local dining hubs and an unoptimized, frustrating experience trying to navigate Pampanga's food map manually.

To resolve these operational, spatial, and data-onboarding structural deficiencies, this study introduces Kanyamanan: A Web-Based Smart Culinary Itinerary and Restaurant Aggregator System. Moving beyond a basic restaurant finder, this project is engineered as a research-grade Decision Support System designed to dynamically coordinate, optimize, and map multi-vendor culinary exploration through a centralized data-ingestion framework. To eliminate merchant operational burdens, the platform deploys a pre-seeded data model populated by public regional tourism listings, allowing the system to be fully functional for tourists out of the box without requiring manual merchant sign-ups, live dashboard management. The architecture consists of a robust administrative portal for data curation, an optional claimable web portal for verified restaurant owners to manage their public data, and a highly responsive consumer web application optimized for desktop and mobile browsers. By integrating custom regional spatial routing logic, predictive crowd density metrics, computer vision-based nutrient analysis (Ando et al., 2021), and an automated dietary budgeting engine, Kanyamanan establishes a highly defensible, realistic, and scalable software model for modern web engineering and intelligent data aggregation in high-density culinary tourism environments.

1.2 Project Context

The system conceptualization of Kanyamanan stems from an investigation into how culinary tourism is managed in Pampanga. The existing process consists of fragmented, self-directed travel planning where tourists rely on isolated data sources. Currently, this practice is performed by manually cross-referencing blog posts, checking standalone social media pages for business hours, using generic mapping software to manually pin addresses, and tracking caloric intake through separate wellness apps. This unintegrated approach introduces severe technical, spatial, and operational inefficiencies. Spatially, generic mapping platforms completely lack the multi-stop routing logic required to cross-examine overlapping merchant operational hours against regional transit delays (Gavalas et al., 2022). Operationally, decentralization

means data remains unsynchronized; dynamic factors such as seasonal menu cost changes, local crowd congestion, and dish allergen contents are hidden from travelers. Consequently, three core groups are heavily affected: culinary tourists who endure exhausting, overlapping transit lines and long queues; local culinary MSMEs who remain digitally invisible due to technical onboarding barriers; and provincial tourism officers who lack centralized, anonymized aggregate metrics to monitor geographic foot traffic trends and optimize future development. Real-world evidence of these operational gaps is regularly observed during weekend holiday surges, where major arterial highways experience massive traffic standstills concentrated around popular local dining hubs. Consumers regularly encounter unexpected restaurant closures, long waiting lines, and menu pricing discrepancies due to outdated public information. A technology-based web intervention is necessary to consolidate these decentralized elements into a unified, informative grid. By aggregating public registries into an open operational framework, software engineering can mitigate regional transit friction, bridge the multi-vendor data gap, and provide automated health metrics without adding technical overhead to local merchants (Fang et al., 2022).

The conceptual foundation of this platform is anchored in its name, *Kanyamanan*, which is a deliberate, linguistic portmanteau of two highly significant Kapampangan terms: *Manyaman*, meaning that the food tastes excellent, delicious, satisfying, or highly pleasing to the palate, and *Kayamanan*, meaning treasure, heritage, or invaluable physical and cultural wealth. In explaining this synthesis, the project defines *Kanyamanan* as "the culinary treasures of Pampanga." It positions gastronomy not simply as a transactional dining activity, but as a living, intangible heritage that must be preserved, celebrated, and made accessible. Through this web ecosystem, traditional recipes, ancestral kitchens, and local food hubs are elevated as regional treasures, linking the sensory delight of *manyaman* dining with the cultural preservation of Pampanga's *kayamanan*.

Kanyamanan stands as a unique intervention in the hospitality technology space by bridging the gap between passive directories and active transaction ecosystems. Unlike generic global mapping directories (e.g., TripAdvisor or Yelp) which treat dining as isolated search queries, this platform is mathematically optimized to handle multi-vendor spatial itineraries mapped with cultural landmarks "along the way" (Gavalas et al., 2022). Its uniqueness is driven by three distinct pillars. First, it features a dual-purpose routing engine that solves the spatial routing problem while simultaneously educating the user about nearby historic Kapampangan landmarks, turning transit times into cultural journeys (Briola et al., 2025). Second, its zero-onboarding hybrid architecture allows tourists to immediately discover pre-seeded restaurants compiled from public registries (Melnick et al., 2022). Third, through active transaction integration, restaurants that choose to register and claim their accounts can unlock a built-in reservation, pre-ordering, and delivery system. This minimizes physical wait times at dining spots, facilitates order planning, and permits local merchants with independent delivery services to safely coordinate shipments to nearby locations. In terms of usability, the platform leverages Progressive Web App (PWA) guidelines to deliver an app-like experience directly within any standard desktop or mobile web browser without requiring users to download native packages. Client-side caching ensures that active itineraries, map directions, contact numbers, and offline tracking remain fully functional even when traversing low-signal areas across remote agricultural and coastal municipalities of Pampanga.

The implementation of *Kanyamanan* is critical to addressing several structural needs within the region. Primarily, there is an urgent need for centralized heritage aggregation. Pampanga's culinary landscape is heavily decentralized, and small-scale, highly authentic ancestral kitchens often operate in digital isolation, lacking the technical resources or literacy to compete with global food franchises (Acuña et al., 2025). This platform serves as an open-access aggregator that lists them alongside major food hubs automatically.

Furthermore, there is a need for logistical optimization. Food tourists face major route uncoordination, leading to local gridlocks. An interactive navigation engine that estimates dynamic ETAs and real-time routes is needed to balance tourist flow across the entire province (Marpaung et al., 2026). Additionally, the system addresses the need for transactional and delivery frameworks; post-pandemic shifts demand that local MSMEs offer frictionless booking. Providing an account-based portal where merchants can toggle pre-orders, table reservations, and delivery services is a critical operational need to sustain local business health. Finally, the project fulfills a need for cultural advocacy. With rapid urbanization, there is a risk of younger generations losing touch with authentic Kapampangan heritage. This study mitigates that risk by gamifying cultural exploration and rewarding users for actively visiting and championing regional culinary spots.

Empirical assessments of regional travel patterns show that a substantial portion of culinary tourists visiting Pampanga are same-day excursionists from adjacent metropolitan regions. Unlike overnight travelers who navigate flexible temporal windows, these day-trippers are strictly bound by sub-24-hour schedules, financial cumulative costs, and the fast physical satiety thresholds associated with the rich, high-density profiles of traditional Kapampangan cuisine. Because existing commercial mapping applications treat food trails as boundless checklists rather than constrained journeys, travelers face logistical over-scheduling, cognitive fatigue, and severe gridlock around major hubs. A specialized web system must therefore offer automated constraint scheduling tailored to these specific localized boundaries.

Statement of the Problem

The main question this study seeks to address is: How can the logistical friction, data fragmentation, and lack of system integration across the decentralized culinary tourism ecosystem of Pampanga be resolved through an open-access web infrastructure?

Specifically, this study aims to answer the following research questions:

- How can the platform resolve the digital isolation of traditional heritage restaurants and ancestral kitchens that currently operate as separate digital entities lacking an aggregator platform for multi-vendor data synchronization?
- What technical interventions are required to mitigate the logistical transit friction encountered by culinary travelers through a live GPS routing engine with real-time ETA tracking spanning the municipalities and component cities of the entire Province of Pampanga?
- How can the spatial routing algorithm be designed to dynamically incorporate notable Kapampangan historical, cultural, and ecological attractions located "along the way" of calculated multi-stop itineraries?
- In what manner can the system integrate a secure, Django-backend account management module for verified merchants to coordinate table reservations, culinary pre-orders, and optional delivery services without complex dashboard overhead?
- How can the platform gamify the preservation of regional culinary arts by engineering a verifiable "Kapampangan Heritage Champion" digital certificate reward system that users can dynamically generate, download, and share online after fulfilling specific regional dining milestones?

1.3 United Nations Sustainable Development Goals (SDGs)

The development and deployment of Kanyamanan directly support global sustainability initiatives, aligning with the institutional research priorities of the University of the Assumption by leveraging software engineering to address socio-economic, health, and cultural heritage challenges.

SDG 3: Good Health and Well-being (Target 3.4)

The platform integrates advanced health informatics to promote long-term well-being and dietary mindfulness among mobile consumers. By deploying a computer vision-based calorie and macronutrient estimator alongside an automated menu ingredient deconstructor, Kanyamanan allows users to process food photographs instantly through a responsive web interface to check estimated nutritional indices against personalized health targets. The system actively flags potential allergen barriers, such as shrimp paste or nuts common in traditional Kapampangan cooking, and identifies structural dietary indicators like high saturated fats. This data-driven health modeling ensures that multi-stop culinary tours remain balanced, supporting user health and well-being directly in the field.

SDG 8: Decent Work and Economic Growth (Target 8.9)

The project establishes a digital framework designed to drive sustainable tourism and boost the market visibility of decentralized culinary MSMEs without creating financial or operational strain. Through its pre-seeded aggregator engine populated via public registries, the system integrates ancestral kitchens and small food hubs into a provincial routing matrix automatically. Local business owners do not need to possess advanced technical literacy, complete mandatory registration, or manage complex dashboards to be discovered by incoming travelers. Furthermore, the optional claimable web portal provides verified merchants with access to anonymous aggregate tourist footprint maps, allowing them to leverage data analytics to support economic growth and operational planning.

SDG 9: Industry, Innovation, and Infrastructure (Target 9.b)

The development and deployment of Kanyamanan directly support SDG 9 by pioneering localized digital innovation and modernizing the technological infrastructure available to the regional hospitality and micro-, small-, and medium-enterprise (MSME) sectors. By constructing an open-access web architecture driven by advanced cloud processing, algorithmic decision support systems, and automated data ingestion, the project establishes a resilient digital public infrastructure. This setup actively drives technological upgrading and domestic innovation within Pampanga's traditional heritage food spaces. It elevates the digital capabilities of local ancestral kitchens and independent venues without demanding complex technical literacy, upfront operational overhead, or expensive native hardware infrastructures from small-scale business owners.

SDG 11: Sustainable Cities and Communities (Target 11.4)

Kapampangan culinary traditions represent a vital component of intangible cultural heritage. Kanyamanan serves as a resilient digital public infrastructure that reinforces the safeguarding and preservation of this distinct legacy. By organizing and categorizing decentralized culinary data and nearby cultural landmarks across the province, the system diverts consumer traffic away from over-congested commercial hubs and routes travelers intentionally toward historical institutions, ancestral kitchens, and localized food spaces (Briola et al., 2025). To drive active community participation, the system features a "Kapampangan Culinary & Cultural Heritage Champion" digital certificate reward module. When travelers complete curated itineraries, verify their physical visits at historical dining establishments via geo-location triggers, and log their culinary journeys, the Django-powered backend generates a beautifully designed, formal digital certificate of achievement. This credential, signed electronically by the administrative tourism portal, recognizes the user's active advocacy for the

preservation of Pampanga's intangible food culture, allowing them to share their achievements across social networks to promote sustainable, community-led heritage tourism (Guevarra & Vergara, 2022).

1.4 Purpose and Description

The primary purpose of this capstone project is to engineer Kanyamanan: an open-access, research-grade Decision Support System that addresses data fragmentation and logistical bottlenecks within the culinary tourism sector of Pampanga. The system operates exclusively within a web environment, accessible across desktop and mobile web browsers without requiring the installation of native applications. While it does not handle direct point-of-sale financial transactions, the software coordinates multi-vendor navigation, table reservations, culinary pre-orders, local delivery setups, dietary tracking, and regional traffic mapping.

The technical architecture is divided into three distinct operational interfaces, driven by a high-performance Django (Python) backend framework which coordinates routing, handles transaction payloads, manages databases, and deploys deep learning scripts. The first interface is the Centralized Data-Ingestion & Django Admin Dashboard. This backend administrative tool allows system administrators to curate master records of restaurants and surrounding cultural attractions across the province. Utilizing Django's secure database model, administrators can pre-seed and partition establishment profiles across various municipalities, complete with standard operating schedules, historical peak occupancy patterns, and verified geographic coordinates.

The second interface is the Claimable Django Merchant Portal, a secure dashboard that allows verified restaurant owners to "claim" their pre-seeded public profiles. Once claimed, merchants can customize their digital menus and toggle active options for table reservations, culinary pre-orders, and optional delivery setups. Django's robust middleware manages these transactional

requests securely, allowing merchants to view active customer booking logs, incoming dish orders, and anonymous aggregate tourist footprint maps to optimize local business operations.

Finally, the Responsive Client Web Application (PWA) serves as the user-facing interface. Designed for both mobile and desktop browsers, this React-rendered frontend connects to the Django API, enabling tourists to generate optimized multi-stop itineraries across Pampanga. Crucially, the app integrates real-time Google Maps-style navigation, tracing live GPS user coordinates along the route and recalculating dynamic ETAs. It highlights historic Kapampangan landmarks situated directly "along the way," displays predictive crowd density charts, manages dietary budget metrics, and features a Django-linked computer vision interface to estimate caloric values. Furthermore, it handles the milestone logging required to award the downloadable "Kapampangan Heritage Champion" certificate of achievement.

1.5 Objectives

The general objective of this study is to design, develop, and evaluate Kanyamanan: A Web-Based Smart Culinary Itinerary and Restaurant Aggregator System. The system serves as a browser-accessible Decision Support System configured to coordinate multi-vendor exploration, automate travel sequence optimization, and deliver health informatics to improve regional tourism efficiency and boost local merchant visibility.

Specifically, the project aims to accomplish the following technical milestones:

1. Design, develop, and implement the core Kanyamanan: A Web-Based Smart Culinary Itinerary and Restaurant Aggregator System utilizing a robust and secure Django (Python) backend database architecture.

2. Develop a Pampanga Provincial Food Trip Web Planner and Algorithmic Routing Engine seamlessly coupled with a Live GPS Map Route Tracker with dynamic ETA tracking, an Along-the-Way Attraction Mapper, an adaptive Highway Traffic Adjuster module, and a Predictive Crowd Density Forecaster analytical framework.
3. Construct an intelligent transactional and health informatics core consisting of a Merchant Reservation, Pre-Order, and Delivery Coordinator Module, an automated Menu Ingredient Deconstructor, a macro budgeting engine, and a mobile-browser-accessible Computer Vision Calorie & Macronutrient Estimator.
4. Implement Intelligent Web-Integrated Assistant and Gamification Modules composed of a preference-based Personal Taste Matcher, a natural language Interactive Travel Assistant Chat interface, and the "Kapampangan Heritage Champion" Verifiable Certificate Achievement Reward generator supported by frontend Progressive Web App (PWA) offline caching routines.
5. Evaluate the technical quality, performance efficiency, and overall user acceptability of the complete integrated web platform across diverse multi-device desktop and mobile web environments in strict accordance with the criteria and quality sub-characteristics defined by the ISO/IEC 25010 Software Quality Model.

1.6 Scope and Delimitation

Scope

The project encompasses the design, development, implementation, testing, and evaluation of a web-based aggregator, transaction manager, and itinerary planning platform configured for the culinary tourism sector across the entire Province of Pampanga, accessible across desktop and mobile web browsers. The system architecture follows a three-tier model composed of a centralized Django backend for administrators, a claimable Django merchant portal for verified restaurant owners to manage bookings, and a highly responsive client PWA application for tourists.

Functional modules are restricted to:

- The pre-seeded multi-vendor database engine spanning all municipalities and component cities of Pampanga.
- The Live GPS Navigation Map Route Tracker with step-by-step path tracing and dynamic ETA calculations (Marron et al., 2025).
- The algorithmic multi-stop trip planner featuring parameter bounding that suggests historical Kapampangan attractions along the way of computed dining paths (Gavalas et al., 2022; Briola et al., 2025).
- The Reservation, Pre-Ordering, and Delivery setup coordinator for claimed merchant accounts.
- The "Kapampangan Heritage Champion" Certificate Generation Engine.
- Historical peak crowd forecasting, mobile computer vision models for food image analysis, menu ingredient deconstruction, and natural language travel chat assistant modules.

Delimitations

- No Direct Payment Gateway Processing: While the system securely coordinates table reservations, pre-orders, and delivery configurations between users and claimed merchants, it does not process monetary payments. All financial transactions are finalized off-platform (e.g., via

Cash on Delivery, GCash/bank transfer directly to the merchant, or physical pay-at-counter dining) to avoid heavy financial data overhead and PCI-compliance boundaries.

- Zero Mandatory Merchant Onboarding: The website does not mandate active merchant registration for core navigation. Tourists can still route to and discover pre-seeded restaurant profiles, although pre-ordering, reservations, and delivery capabilities will remain locked for that venue until the verified business owner claims their profile.
- No Native App Distribution: The platform is exclusively a web-based application built using React and Django (Python); it will not be developed for or published on native application stores (Google Play Store/Apple App Store).

CHAPTER II: REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Foreign Literature

The rapid digitalization of the global tourism and hospitality sectors has precipitated a fundamental structural shift from static digital directories to dynamic, highly integrated ecosystem grids. A primary focal point in contemporary computational tourism literature is the resolution of the Personalized Tourist Trip Design Problem (PTTDP), a complex multi-objective optimization challenge where travel itineraries must continuously balance geographic distances, temporal constraints, and user-specific preferences (Ahmad & Halim, 2023; Marpaung et al., 2026). In the era of "Tourism 4.0," conventional navigation services, which provide foundational point-to-point routing, are structurally inadequate for culinary travelers who require multi-stop itineraries bounded by overlapping vendor operational hours, dynamic traffic densities, and specific dining durations (Gavalas et al., 2022). The literature emphasizes that tourists frequently operate under strict time budgets and demonstrate a high aversion to logistical friction (Hall & Gössling, 2022). For instance, complex transit transfers, unpredictable waiting queues, and the nuances of public transit systems in unfamiliar destinations are perceived as significant deterrents that diminish the experiential value of a location (Hall & Gössling, 2022). Field studies highlight that travelers seek to maximize the time spent at a destination by minimizing transit time, making the integration of advanced Information and Communication Technology (ICT) tools indispensable for modern travel planning (World Tourism Organization [UNWTO], 2023).

To mitigate these spatial inefficiencies, theoretical computational models have increasingly integrated algorithmic metaheuristics into web-based geographic information systems. Advanced mathematical approaches utilizing the Adaptive Neighborhood Simulated Annealing (ANSA) algorithm, which is a sophisticated enhancement of the traditional Simulated Annealing heuristic, have been proposed to solve extended models of the Team Orienteering Problem with Time Windows (TOPTW). Such algorithmic frameworks dynamically suggest

locations and activities to maximize a traveler's accumulated satisfaction score while adhering strictly to mandatory geographic and temporal constraints, such as a restaurant's opening and closing hours. The literature posits that embedding these metaheuristics directly within responsive web frameworks allows for the rapid, real-time recalculation of travel sequences, ensuring that tourists are automatically rerouted away from congested corridors toward optimized, secondary tourist destinations. This shift from passive mapping to active, algorithmic tour planning is foundational for high-density tourist regions where optimal spatial distribution is required to prevent localized overcrowding.

Parallel to the advancements in spatial routing algorithms, foreign literature extensively documents the integration of artificial intelligence (AI) and machine learning into health informatics, clinical nutrition, and dietetics (Panayotova, 2025). As consumers increasingly demand personalized wellness tracking capabilities integrated seamlessly into their mobile and web environments, nutrition science has shifted aggressively toward automated dietary assessment (Fang et al., 2022). The literature highlights the development of image-based AI tools, such as the goFOOD 2.0 system, which utilize advanced computer vision algorithms and deep neural networks to identify food items, estimate volumetric portion sizes, and calculate caloric and macronutrient indices directly from digital photographs (Ando et al., 2021; Panayotova, 2025). These automated AI frameworks represent a monumental leap over traditional manual logging methods, which the literature heavily criticizes as notoriously prone to human error, recall bias, and significant underreporting (Panayotova, 2025).

Furthermore, deep learning architectures are increasingly deployed within dietary assessment tools to deconstruct menu ingredients, flag potential lifestyle constraints, and identify allergen barriers (e.g., high sodium levels, saturated fats, or specific nut allergies) against personalized physiological profiles (Thierry et al., 2021). The literature also explores the use of large language models (LLMs) and conversational agents to provide real-time nutritional coaching, noting that while

LLMs demonstrate competitive accuracy in identifying calories and proteins, they occasionally exhibit error rates when generating complex, disease-specific meal plans, highlighting the necessity for human-in-the-loop validation or strict algorithmic bounding (Panayotova, 2025).

To support the continuous ingestion of data across decentralized business nodes without imposing manual maintenance burdens, foreign literature advocates for the deployment of centralized aggregator models (Melnick et al., 2022). Rather than forcing separate merchants to perform manual updates, enterprise web aggregation relies on pre-seeded data modeling and public information structures to ensure immediate, out-of-the-box system utility (Melnick et al., 2022). By cleanly partitioning database records, web frameworks can successfully establish public multi-tenant directory maps that abstract architectural complexities away from the end-user, ensuring secure data retrieval and baseline service accessibility across independent establishments (Melnick et al., 2022).

Furthermore, contemporary hospitality literature emphasizes a strict operational distinction between overnight tourists and same-day visitors or excursionists (Hall & Gössling, 2022). While overnight tourists operate under flexible temporal boundaries, same-day excursionists face rigid sub-24-hour time constraints, making them highly sensitive to transit delays and queue friction (Hall & Gössling, 2022). Computational modeling in the Tourist Trip Design Problem (TTDP) highlights that when travelers are bound by tight time budgets and physical satiety limits, the optimization algorithm must enforce a bounded stop ceiling—frequently restricting calculations to a maximum of two or three highly curated locations per journey (Ahmad & Halim, 2023). Metaheuristic frameworks that allow users to input a maximum stop threshold prevent cognitive fatigue and operational over-scheduling, ensuring that the computed sequential paths remain realistic, mathematically sound, and economically viable for short-duration travel footprints (Marpaung et al., 2026).

Summary Statement

The reviewed foreign literature demonstrates that the convergence of algorithmic spatial optimization, computer vision-based nutritional assessment, and multi-tenant data ingestion provides a robust theoretical foundation for modern tourism platforms. The extensive literature underscores the absolute necessity of integrating real-time logistical problem-solving with automated health informatics to overcome the limitations of static directory services. Furthermore, safeguarding these complex web ecosystems through robust, client-side data boundaries is essential for protecting user privacy and ensuring the operational integrity of decentralized culinary aggregator systems.

2.2 Local Literature

Within the Philippine context, the national thrust toward digital governance, smart tourism, and the modernization of the micro-, small-, and medium-enterprise (MSME) sector has accelerated the necessity for localized, technology-driven frameworks. The Philippine Development Plan consistently identifies digital transformation as a fundamental strategy for improving regional economic growth, enhancing citizen engagement, and elevating the global competitiveness of local heritage industries. However, the digital transition of the culinary tourism sector, particularly in historically rich and high-density gastronomic hubs like the Province of Pampanga, remains heavily fragmented. Existing regional tourism infrastructure often relies on uncoordinated, paper-based promotional materials or decentralized, standalone social media pages, resulting in a severe lack of synchronized operational data for incoming tourists. The local literature argues that establishing centralized aggregator platforms can significantly reduce this digital isolation, systematically channeling consumer traffic toward historically significant ancestral kitchens, thereby

protecting localized food spaces from unsustainable urban congestion while boosting local economies.

A critical component governing the implementation of any centralized data aggregator system in the Philippines is the strict adherence to the legal and regulatory frameworks surrounding data privacy and digital security. Republic Act No. 10173, comprehensively known as the Data Privacy Act of 2012 (DPA), establishes stringent, non-negotiable operational guidelines for the collection, processing, storage, and cross-border transfer of personal and sensitive information. The legislation applies broadly to any natural or juridical person engaged in personal information processing, mandating that technological platforms uphold the fundamental rights of data subjects, which explicitly include the rights to transparency, legitimate purpose, and data proportionality. The DPA ensures that individuals maintain granular control over their personal information, empowering them to dictate how their private data is accessed and utilized by both government authorities and private sector entities.

With the integration of transactional and operational configurations, specifically the platform's table reservation, culinary pre-ordering, and delivery services, adherence to the Data Privacy Act of 2012 (RA 10173) becomes structurally critical. Because the platform collects sensitive user geolocation coordinates for delivery mapping, personal contact details for booking verifications, and historical dining logs, the system architecture must deploy strict privacy-by-design standards.

In accordance with the National Privacy Commission guidelines for web aggregators, Kanyamanan minimizes personal data collection to the bare minimum required to execute active reservations or coordinate deliveries. The Django backend implements robust cryptographic security measures, utilizing role-based access control (RBAC) to ensure that customer address details and contact information are strictly visible to the specific merchant fulfilling the order and are programmatically masked or purged upon transaction completion. This

ensures that tourist movements, dining habits, and physical locations are safeguarded, maintaining absolute compliance with the confidentiality mandates of Section 8 of RA 10173.

For web-based systems that incorporate user profiling, customized dietary tracking, geographical location monitoring, and merchant verification portals, compliance with RA 10173 is not merely an ethical consideration but a strict legal prerequisite. Local legal and IT literature emphasizes that businesses handling personal data must navigate complex requirements, most notably the mandate to obtain valid, informed, and explicit consent before executing any data processing routines. Furthermore, Section 8 of the DPA strictly mandates that personal information controllers and processors must "ensure at all times the confidentiality of any personal information that comes to its knowledge and possession". This specific legislative parameter directly influences software system architecture, requiring the integration of robust security protocols, such as role-based access control (RBAC), database encryption, and secure communication layers (e.g., HTTPS/TLS) to safeguard both the behavioral data of culinary tourists and the proprietary operational data of verified restaurant claimants.

The local literature also highlights the severe legal, financial, and reputational ramifications of non-compliance, which include substantial monetary penalties and institutional damage. Consequently, local researchers advocate for proactive privacy-by-design principles during the software engineering lifecycle. This methodology entails minimizing the collection of identifiable consumer data wherever anonymized aggregate metrics can suffice. For instance, when supplying local tourism officers or verified merchants with footprint analyses, predictive crowd density charts, or overall regional engagement metrics, the system must process this data anonymously. By doing so, the platform ensures that geographic foot traffic trends and culinary popularity indices are derived without compromising the identity, specific movement patterns, or personal health targets of individual users. Furthermore, while the DPA provides certain

exemptions, such as information about individuals who are or were officers of government institutions, the overarching mandate for commercial web platforms remains stringently focused on consumer protection.

Summary Statement

The reviewed local literature emphasizes the critical intersection between regional digital transformation initiatives and the stringent data privacy regulations established in the Philippines. The synthesis of these sources confirms that while the aggregation of culinary MSMEs into centralized web platforms is absolutely essential for sustainable regional tourism growth, such software architectures must be fundamentally anchored in the privacy-by-design mandates of Republic Act No. 10173. Adherence to these legal frameworks ensures the secure, confidential, and legally compliant processing of stakeholder information, fostering trust among tourists and local merchants alike. Upholding these digital benchmarks directly advances local industrial innovation by creating a resilient web infrastructure for historically underserved merchants.

2.3 Foreign Studies

Empirical investigations, quantitative modeling, and experimental studies conducted internationally provide critical operational benchmarks for the algorithmic and artificial intelligence components required in dynamic, multi-vendor web systems. A highly relevant domain of foreign research involves the practical application of advanced metaheuristics to successfully solve the Personalized Tourist Trip Design Problem (PTTDP). In a pivotal empirical study by Marpaung et al. (2026), researchers engineered a comprehensive Spatial Decision Support System (SDSS) that seamlessly fused the Ant Colony Optimization (ACO) algorithm with a Web-based Geographic Information System (WebGIS) to optimize tourist routing in North Sumatra. The study recognized that the transition toward "Tourism 4.0" redefined travel planning into a multifaceted optimization challenge that conventional navigation services were unequipped to handle. The quantitative investigation utilized a sophisticated weighted-sum

scalarization technique to harmonize divergent multi-objective goals: maximizing the attraction scores of destination points while simultaneously minimizing both travel distance and transit duration.

The empirical outcomes of the Marpaung et al. (2026) study revealed profound efficiency gains. When benchmarked directly against traditional navigational routing mechanisms, the ACO-WebGIS framework achieved a remarkable 17.84% reduction in total travel distance (reducing typical travel from 18.5 km to 15.2 km) and a 17.24% reduction in estimated travel time (reducing duration from 145 minutes to 120 minutes). Concurrently, the algorithmic optimization increased the number of points of interest (POIs) successfully visited by 42.85%, raising the average attraction score by 22.37%. Crucially for web application development, the system maintained a highly responsive computational latency of merely 1.45 seconds. The researchers concluded that the seamless coupling of algorithmic metaheuristics with interactive spatial web mapping empowers tourists to evaluate computed optimal routes directly on a browser interface, thoroughly overcoming the inherent limitations of static navigation tools. A separate seminal study by Gavalas et al. (2022) on the eCOMPASS multimodal tourist tour planner corroborates these spatial findings, demonstrating that algorithmically generated travel sequences significantly alleviate the cognitive and logistical burdens placed on tourists navigating unfamiliar metropolitan transit networks and highly congested tourist corridors.

In the realm of health informatics and dietary tracking, extensive foreign experimental studies have evaluated the precise efficacy of artificial intelligence. Panayotova (2025) conducted a comprehensive systematic review analyzing peer-reviewed publications across PubMed, Scopus, and Web of Science regarding AI in nutrition and dietetics. The study categorized findings into key thematic areas, revealing that image-based AI dietary tools, such as the goFOOD 2.0 system, utilize deep neural networks and computer vision to achieve moderate to high agreement with ground-truth data and registered dietitian assessments under real-world clinical conditions. AI-driven models

successfully estimate caloric and nutritional intake by leveraging computer vision and natural language processing, significantly enhancing dietary tracking accuracy and personalizing disease-specific nutrition management. However, Panayotova's (2025) research also warned of ongoing challenges, specifically regarding model transparency, the ethical use of health data, and the limitations of these models when applied to low-resource settings or highly diverse, non-Western diets.

Further advancing the mechanics of computer vision in nutrition, Ando et al. (2021) investigated volumetric estimation techniques for complex food dishes utilizing advanced depth sensors integrated with smartphones. Their experimental pipeline achieved a robust classification accuracy of 95% and maintained a mean absolute percentage error (MAPE) for volume prediction of 16% on completely untrained object datasets. Seeking to eliminate the hardware dependency on depth sensors, other experimental configurations have integrated semantic segmentation and point cloud modeling using standard 2D imagery (Thierry et al., 2021). These approaches have achieved mean squared errors (MSE) as low as 5.61 and mean absolute errors (MAE) of 1.07, proving highly viable for practical deployment in standard mobile web browsers. Studies focusing on specific geometric food containers, such as dining bowls, proved that hybrid computational algorithms frequently outperform independent human estimators in calculating food volumes and subsequent caloric loads. Additionally, approaches integrating the Enhanced Squirrel Search Algorithm (ESSA) and Convolutional Neural Networks (CNN) for multi-view image processing have achieved remarkable precision, yielding a mean relative error of only 3.95% in calorie prediction on benchmark datasets like Nutrition5k (Thierry et al., 2021).

In the domain of database management and zero-onboarding architectures, an empirical study by Melnick et al. (2022) evaluated the structural design of web aggregator grids configured for small-scale decentralized merchants. The quantitative investigation revealed that forcing independent

MSME operators to manage active data dashboards results in high system abandonment rates due to technical literacy barriers and kitchen operational overhead. To circumvent this bottleneck, the researchers validated a pre-seeded data ingestion framework that uses public registry parameters to build fully operational vendor profiles automatically. The system partitions records securely while leaving profiles open to be optionally claimed by verified owners later. The study concluded that pre-seeded multi-vendor aggregation guarantees a high volume of active points of interest immediately, making it the ideal architectural blueprint for regional digital ecosystems where mandatory merchant onboarding is logistically impossible.

Summary Statement

Foreign studies rigorously validate the immense technical feasibility and operational superiority of integrating metaheuristic algorithms, such as Ant Colony Optimization, into web-based spatial routing systems to solve complex travel logistics. Concurrently, experimental research in computer vision and semantic segmentation demonstrates that AI can accurately estimate nutritional volumes from standard digital images without requiring specialized depth hardware. Furthermore, data science research into multi-vendor aggregation underscores the structural viability of pre-seeded data parsing. These global empirical findings provide the exact computational blueprints necessary for constructing intelligent, secure, and highly optimized web aggregator platforms.

2.4 Local Studies

Within the Philippine educational and industrial landscape, scholarly investigations conducted by local IT researchers provide critical technical baselines for localized web integration, system architecture adoption, and user experience metrics. Preparing enterprise systems necessitates evaluating how close-to-home research addresses multi-vendor tracking, spatial navigation, automated user assistance, and regional information curation.

Guevarra and Vergara (2022) developed an interactive platform titled *I-Tribe: Aeta Tribe Culture Preservation Application with Registry and Documentation* designed to track population records and preserve the history and customs of indigenous communities in Mawacat, Floridablanca, Pampanga. Using a qualitative approach and a traditional five-phase Waterfall methodology spanning requirements, design, implementation, verification, and maintenance, the researchers engineered a web system utilizing PHP, HTML, CSS, and Figma prototyping software integrated with a structured database and administrative portal. Evaluation results from 33 participants and IT specialists demonstrated that the system was simple to operate, maintained secure data storage, and achieved high usability ratings, establishing an institutional precedent for partitioning regional profiles without compromising data integrity.

Bunag and Tuazon (2022) engineered *A Web-based Dynamic Chart Budget Tracker Application for Enterprises Powered by Complex Voice* to minimize manual typing errors and prevent financial overspending through hands-free interaction. Utilizing a quantitative, descriptive-developmental approach with data from 90 small-to-medium enterprise operators in Pampanga, the study implemented a system following the sequential Waterfall methodology. The application's frontend interface was constructed using React JavaScript and Visual Studio Code, integrating Speechly voice-command software within an Input-Process-Output framework to translate spoken phrases into a centralized database that outputs dynamic graphical charts. Implementation results showed that 90% of respondents considered the interface intuitive, user-friendly, and effective for organizing financial cash flows, proving that automated voice-tracking reduces tracking friction.

Marron et al. (2025) addressed the lack of centralized public transit records in Central Luzon by developing *Mappa Terminal: A Transport Station Map App with AI Chatbot Integration in City of San Fernando Pampanga*. Following an iterative Agile software development methodology and a descriptive-developmental mixed-methods design, the developers gathered data

from 100 commuter respondents and local transit terminal personnel. The resulting Android-based mobile application combined interactive 2D and 3D mapping grids driven by the GraphHopper routing engine with a natural language processing core powered by the Gemini API to resolve commuter queries regarding fares, terminals, and schedules. Quantitative testing based on the Technology Acceptance Model (TAM) yielded a very high Grand Mean of 4.57, with expert validations confirming strong technical performance in functional suitability and reliability while successfully minimizing commuter stress.

Acuña et al. (2025) launched *FARMARKET: A Digital Marketplace System For Local Young Farmers In Municipalities of Mexico, Lubao, and Sta. Ana Pampanga* aimed at eliminating middleman dependencies and expanding market reach for local producers. Grounded in the Technology Acceptance Model (TAM), this descriptive-developmental study gathered mixed-methods data through user acceptance testing with 70 young farmers and 5 IT professionals. The platform was built using an Agile development lifecycle to integrate key features like secure email verification, direct farmer-to-consumer messaging, order management tables, a report produce option, and a comprehensive farmer dashboard generating analytical sales charts. Usability evaluations yielded high grand means of 3.36 for both perceived usefulness and ease of use, showing that the system's intuitive design effectively accommodates agricultural stakeholders with varying digital literacy levels.

Briola et al. (2025) developed *EcoLakbay: A Sustainable Tourism Web Platform for Pampanga* to guide regional travelers toward eco-conscious decisions and environmentally responsible travel behavior. The study utilized an exploratory sequential mixed-methods design, collecting system requirements and usability evaluations from adult travelers, local hospitality business owners, municipal tourism officers, and expert validators. Developed using iterative sprint cycles within an Agile framework combining Scrum and Kanban, the web application implemented an AI travel assistant for personalized path planning, a partner destination registration module, a community sharing wall, and a Carbon

Footprint Calculator to estimate travel emissions. Evaluation using the System Usability Scale (SUS) achieved a grand mean of 3.46, signifying very high acceptability, while an ISO/IEC 25010 expert checklist validated a high grand mean of 3.57 across functionality, security, and portability attributes.

Summary Statement

The reviewed local studies demonstrate that the transition from manual, uncoordinated record-keeping to centralized web-based management systems exponentially improves data accuracy and user performance across the public, agricultural, and tourism sectors of the Philippines. These local capstone frameworks confirm that combining automated database engines, metaheuristic mapping paths, and natural language interfaces into intuitive browser environments effectively abstracts complex software logic away from the user. By studying the empirical outcomes of these institutional systems, a solid software engineering foundation is established. This foundation supports the integration of multi-vendor aggregation matrices, automated macro budgeting thresholds, and progressive cross-municipality travel paths into a unified decision support ecosystem tailored specifically for the Province of Pampanga.

2.5 Theoretical and Conceptual Framework

The engineering, deployment, and evaluation of the proposed Kanyamanan web-based system are deeply anchored in internationally recognized software engineering frameworks and rigorous developmental life cycle models. These theoretical constructs ensure that the resulting application is systematically built, structurally sound, legally compliant, and highly capable of functioning reliably within dynamic, high-traffic digital environments.

ISO/IEC 25010 Software Quality Model

To guarantee that the web platform meets enterprise-grade operational standards, the study adopts the ISO/IEC 25010 Software Quality Model as its primary evaluative framework. Recognized globally by the International

Organization for Standardization, the ISO/IEC 25010 model replaces the older ISO 9126 standard and provides a comprehensive, multi-dimensional taxonomy for assessing software product quality. The framework categorizes system quality into eight primary characteristics, each containing specific sub-characteristics that directly dictate architectural and engineering decisions throughout the project:

1. **Functional Suitability:** This measures the degree to which the software provides functions that meet stated and implied needs when used under specified conditions. For the proposed platform, this dimension rigorously evaluates whether the algorithmic routing engine accurately calculates spatial distances and traffic modifiers, whether the computer vision module successfully outputs macronutrient estimations from user-uploaded images, and whether the administrative backend correctly partitions municipality data.
2. **Performance Efficiency:** This assesses the system's performance relative to the amount of resources utilized under stated conditions. Given the findings from studies like Marpaung et al. (2026), the system must ensure that the metaheuristic routing algorithms process multi-stop itineraries with minimal latency (e.g., generating a route in under two seconds) while optimizing client-side memory consumption during heavy image processing tasks.
3. **Compatibility:** This evaluates the system's ability to exchange information with other products or systems (interoperability) and perform its required functions while sharing a common hardware or software environment (co-existence). The platform must seamlessly interface with third-party web mapping APIs (such as Google Maps API or OpenStreetMap) and external weather/traffic data endpoints without conflict or data loss.
4. **Usability:** This determines the degree to which a product can be used by specified users to achieve quantified objectives with effectiveness, efficiency, and satisfaction. Given that the system targets diverse demographics, ranging from international culinary tourists to local heritage

food vendors, the user interface must possess high operability, learnability, and robust user error protection, avoiding the complex onboarding barriers typical of legacy enterprise systems.

5. **Reliability:** This measures the system's capability to maintain a specified level of performance over a given period. Sub-characteristics include maturity, availability, fault tolerance, and recoverability. The platform's centralized database must prevent data corruption during simultaneous user queries and maintain extremely high uptime during peak weekend culinary tourism surges.
6. **Security:** In absolute accordance with the Philippine Data Privacy Act (RA 10173), the system enforces rigorous data protection protocols. The Django backend provides built-in, automated defenses against common web vulnerabilities, including SQL injection, cross-site scripting (XSS), and cross-site request forgery (CSRF). Secure hashing algorithms are used to store customer and merchant passwords, and role-based access control (RBAC) ensures that customer reservation, delivery, and route logging data are securely partitioned and inaccessible to unauthorized actors.
7. **Maintainability:** This reflects the degree of effectiveness and efficiency with which the software can be modified to improve it, correct it, or adapt it to environmental changes. The web architecture must be highly modular and reusable, allowing future developers to update nutritional databases, refine the computer vision models, or adjust algorithmic routing parameters without requiring total system downtime.
8. **Portability:** Engineered as a web-based Progressive Web App (PWA) using React for the presentation layer and Django for the API, the system demonstrates high adaptability. It ensures that the live GPS map navigation, ETA calculations, and interactive attraction maps render flawlessly across various mobile and desktop browsers (Chrome, Safari, Edge) without requiring native rewrite.

System Development Life Cycle Framework

The engineering and deployment of the web platform strictly adhere to the Agile (Scrum) software development methodology. Unlike rigid, traditional linear frameworks such as the Waterfall model, Agile is highly iterative and incremental, emphasizing continuous stakeholder feedback, rapid adaptation to technical hurdles, and the frequent delivery of functional software modules. Given the highly complex, multi-layered nature of an aggregator system featuring AI and routing algorithms, Agile mitigates project risk by breaking development into manageable cycles.

The Agile model comprises several cyclical phases executed within designated timeboxes known as "sprints." During the Plan Phase, the engineering team identifies core system requirements, precisely defining the parameters of the spatial routing algorithms, establishing mapping API integration limits, and compiling the initial pre-seeded database of Pampanga's culinary establishments using public tourism registries. The Design Phase translates these requirements into highly detailed structural blueprints, formulating the relational database schema, mapping the three-tier system architecture, and drafting the responsive User Interface (UI) and User Experience (UX) wireframes for both desktop and mobile endpoints.

The Develop Phase entails the rigorous, hands-on construction of the system. This involves coding the backend API, implementing the algorithmic traffic-adjuster logic, training the vision engine for food classification, configuring the intelligent taste-matching modules, designing the natural language assistant interface, and establishing the PWA caching protocols. Following this, the Test Phase rigorously subjects the system to unit testing, integration testing, and API stress testing. This phase is critical for verifying that the algorithmic routing latency remains low (mirroring the 1.45-second latency achieved in previous ACO studies) and that the computer vision module's nutritional estimates fall within an acceptable margin of error. Upon successful testing, the Release Phase deploys the functional modules to a staging or production web server. Finally, the

Feedback Phase integrates continuous User Acceptance Testing (UAT) evaluations, allowing for rapid codebase refinements based on actual user interaction, ensuring the software continually aligns with the ISO/IEC 25010 standards.

Conceptual Framework

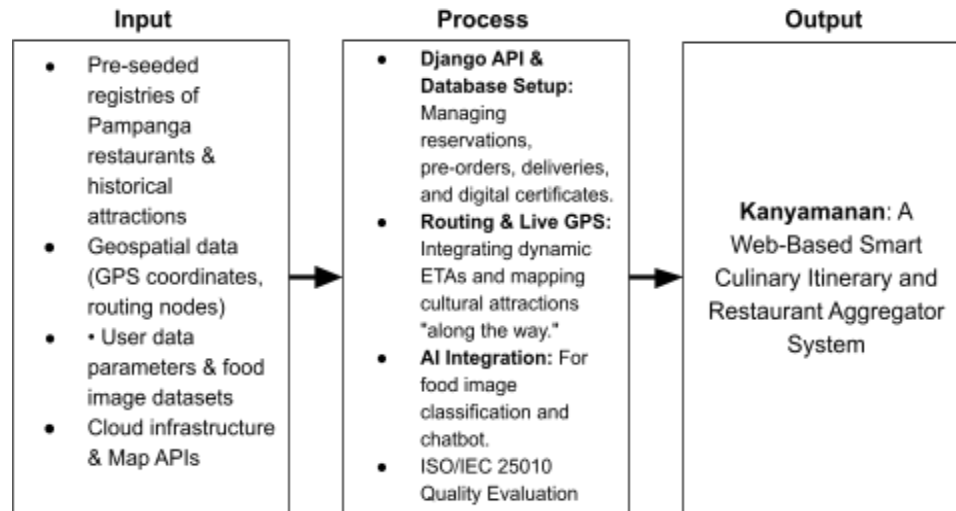


Figure 2.3: IPO Framework

To synthesize the logical flow of the project's development operations, the Input-Process-Output (IPO) model is adopted as the overarching conceptual framework. This model systematically defines the transition of raw data and requirements into a functional, highly optimized digital solution.

The **Input phase** aggregates the raw materials and foundational data required for system execution. These inputs encompass public tourism registries, accurate geographic coordinates of culinary establishments, hardware and software development resources (including cloud infrastructure and mapping APIs), established food image datasets for training the nutritional vision model, and the functional requirements derived from the analysis of the localized culinary tourism sector's friction points.

The **Process phase** represents the transformative computational and engineering actions governed by the Agile methodology. This systematic phase includes the execution of deep requirement analyses, the formulation of complex database schemas, the coding of the algorithmic multi-stop trip planner, the implementation of predictive crowd density forecasting logic, the integration of PWA offline caching techniques, and the rigorous execution of ISO/IEC 25010 quality evaluation testing.

The resulting **Output** is the fully functional Kanyamanan: A Web-Based Smart Culinary Itinerary and Restaurant Aggregator System. This final product serves as an integrated digital infrastructure that successfully coordinates cross-municipality routing, automates nutritional data tracking, and provides a unified, highly visible digital presence for local MSMEs without mandating technical operational overhead.

2.6 Synthesis

The comprehensive review of related literature, empirical studies, and legislative frameworks establishes a clear, multi-disciplinary imperative for the development of an integrated, web-based culinary aggregator system in the Province of Pampanga. An analysis of global tourism mechanics reveals that modern consumers heavily prioritize authentic, localized experiences but are frequently deterred by the severe logistical friction associated with the Personalized Tourist Trip Design Problem (PTTDP). Foreign literature unequivocally demonstrates that integrating advanced algorithmic metaheuristics, such as Ant Colony Optimization or Simulated Annealing, directly into web-based geographic information systems drastically reduces transit times, minimizes travel distances, and significantly enhances the volume of points of interest successfully visited. These algorithmic blueprints validate the absolute necessity and technical feasibility of constructing the Provincial Food Trip Web Planner, Highway Traffic Adjuster, and Predictive Crowd Density Forecaster

modules, ensuring tourists are dynamically routed away from congested provincial corridors and overcrowding patterns.

Simultaneously, the integration of artificial intelligence into health informatics has fundamentally transformed the accuracy and feasibility of dietary tracking. Foreign studies confirm that advanced computer vision algorithms and semantic segmentation models can accurately estimate caloric and macronutrient indices from digital imagery with low margins of error, even eliminating the need for specialized depth-sensor hardware. By synthesizing these highly accurate AI-driven dietary assessment techniques with traditional menu ingredient parsing, the proposed platform is uniquely positioned to offer a highly robust, automated intelligent Health Informatics core. This configuration ensures that an automated Menu Ingredient Deconstructor, a dietary macro budgeting engine, and a Computer Vision Calorie & Macronutrient Estimator work in unison to preserve continuous user dietary mindfulness and actively flag allergen barriers without interruption during extensive culinary tours.

Crucially, local literature firmly grounds these advanced technological capabilities within the specific operational realities, economic goals, and regulatory frameworks of the Philippines. Local empirical studies underscore that the transition from manual, fragmented coordination to centralized web-based management systems exponentially improves data accuracy, standardizes institutional knowledge, and drastically improves operational efficiency. This transition directly supports United Nations Sustainable Development Goal 9 (Industry, Innovation, and Infrastructure) by providing a resilient digital public infrastructure that drives technological upgrading and domestic innovation within Pampanga's traditional heritage food spaces. It effectively modernizes the infrastructure available to the regional micro-, small-, and medium-enterprise (MSME) sector without demanding complex technical literacy or upfront operational overhead from small-scale business owners. However, this vital digital transformation is strictly governed by the mandates of the Data Privacy Act of 2012 (RA 10173). Deep research into local privacy awareness necessitates

that the system be engineered with proactive privacy-by-design principles by utilizing secure pre-seeded data architectures, mandatory consent protocols, and anonymized aggregate analytics. This approach systematically abstracts complex legal compliance burdens away from both the end-user and the local merchant.

The extensive synthesis of these technological, legal, and operational domains reveals a profound research gap: while generic mapping applications, standalone dietary trackers, and isolated social media directories currently exist, there is absolutely no unified, algorithmic web ecosystem tailored to harmonize the spatial, nutritional, and vendor-visibility constraints specific to the high-density culinary tourism sector of Pampanga. Kanyamanan: A Web-Based Smart Culinary Itinerary and Restaurant Aggregator System directly and decisively addresses this gap.

By fusing custom spatial routing, AI-driven nutritional estimations, and rigorous RA 10173 privacy compliance into a Django-powered web architecture, the system introduces a specialized optimization matrix tailored for culinary travelers. It moves beyond static directory lookups by integrating live GPS tracking with dynamic ETA updates, showcasing Kapampangan historical attractions along the way, and enabling a secured reservation, pre-ordering, and delivery setup for claimed merchant accounts.

Furthermore, by incorporating a "Kapampangan Heritage Champion" digital certificate reward system, the ecosystem gamifies cultural advocacy, converting standard food trips into structured achievements. Systematically built using the Agile methodology and evaluated under the eight characteristics of the ISO/IEC 25010 Software Quality Model, this multi-tiered architecture provides a highly defensible, structurally sound, and deeply innovative solution to the complex localized challenges of experiential culinary travel.

CHAPTER III: METHODOLOGY

3.1 Methods of Research

The study used a Quantitative Research Design as its primary research approach. Quantitative Research Design served as a systematic research approach for collecting, analyzing, and interpreting numerical data to evaluate the proposed system objectively. This approach enabled the researchers to measure the quality, usability, functionality, and overall acceptability of the proposed KANYAMANAN platform using standardized evaluation instruments and statistical analysis. This method was considered appropriate for the study because it focused on creating and validating a functional web-based platform that integrated culinary itinerary optimization, health informatics, and restaurant aggregation for culinary tourists in Pampanga. The Quantitative Research Design process consisted of four major phases: analysis, design, development, and evaluation. During the analysis phase, the proponents identified the needs and requirements of the target users through surveys and document analysis. The information gathered during this phase served as the basis for determining the system requirements and identifying the existing challenges encountered by culinary tourists, local culinary MSME operators, and the Provincial Tourism Office.

During the design phase, the proponents developed the system architecture, database structure, user interface prototypes, process models, and other design specifications necessary for the development of the proposed platform. These design outputs provided a clear blueprint for implementing the system and ensured that both functional and non-functional requirements were addressed.

The development phase involved the implementation of the proposed system using appropriate web technologies, programming languages, frameworks, and development tools. The proponents integrated the itinerary optimization algorithms, health informatics module, restaurant aggregation features, and other functional components into a unified web-based platform. Continuous testing and debugging activities were conducted throughout the development process to verify that each module functioned according to the specified requirements.

The evaluation phase focused on assessing the developed system using the ISO/IEC 25010 Software Product Quality Model. The evaluation was conducted with the participation of Information Technology specialists, local culinary MSME operators, culinary tourists and same-day excursionists, and representatives from the Provincial Tourism Office. Their assessments determined the quality of the proposed platform in terms of functionality, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability. Specifically, the evaluation measured the functional suitability and usability of the Django-managed reservation, pre-ordering, and delivery configurations, the real-time accuracy of the live GPS routing and dynamic ETA tracing, and the visual appeal and verification security of the digital certificate reward system. The results of the evaluation provided the basis for identifying possible improvements before the implementation of the proposed system.

The Quantitative Research Design ensured that the proposed KANYAMANAN platform was systematically designed, developed, and evaluated using a structured research process. Furthermore, the methodology supported the achievement of the study's objectives by providing a comprehensive framework for developing a user-centered, technically sound, and reliable web-based culinary tourism platform.

3.2 Systems Development Life Cycle

This section aimed to describe the software development methodology used in developing the proposed system. The Software Development Life Cycle (SDLC) served as the framework that guided the development process from planning to implementation. Common SDLC models included Waterfall, Agile, RAD, and V-Model. Among these models, the Agile Model was selected because it was appropriate for the project's scope, team size, timeline, and complexity. This section explained why the chosen model was suitable and how its phases were applied in the development of the system.

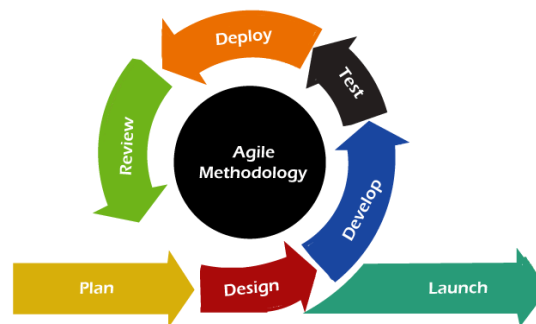


Figure 3.1: Agile Model of System Development

The Agile Model, as shown in Figure 3.1, was an iterative and incremental software development approach that emphasized flexibility, continuous improvement, user feedback, and the rapid delivery of working software (Project Management Institute, 2025). Unlike traditional linear models, Agile allowed developers to revisit and refine system requirements throughout the development process (Hinderks et al., 2022). This made it highly suitable for IT Capstone Projects where user needs could evolve during development and where continuous validation was necessary to ensure system usability and effectiveness.

The Agile Model was structured as an Input-Process-Output framework. As illustrated in Figure 3.1, project requirements were initiated through a structured Plan Phase (the yellow input arrow), progressed through a cyclical and iterative development loop composed of five core phases (Design → Develop → Test → Deploy → Review), and were formally delivered to stakeholders during the final Launch Phase (the green output arrow).

Plan Phase. The Plan Phase acted as the initial input gateway where the system requirements, user stories, and scoping thresholds were established before work entered the active sprint cycle (Project Management Institute, 2025). During this phase, the proponents conducted interviews and consultations with local culinary stakeholders, same-day excursionists, and the Provincial Tourism Officer to identify the logistical friction, data gaps, and tracking needs of Pampanga's tourism corridor. The gathered requirements were analyzed to define the core features of the Kanyamanan platform, thereby establishing a prioritized product backlog. Key requirements included unbounded provincial spatial mapping, covering all municipalities and component cities of Pampanga, same-day travel ceiling options, such as a two-to-three restaurant stop limit, and local health markers, including Kapampangan-specific allergen identification for ingredients such as shrimp paste and nuts. Furthermore, the requirements-gathering process defined the parameters for live GPS-enabled tracking and real-time route progress with dynamic Google Maps-style estimated time of arrival (ETA) updates. It also established the database structure for cataloging notable Kapampangan historical, cultural, and ecological attractions situated directly along calculated transit routes. In addition, the system requirements incorporated merchant configuration settings that supported table reservations, culinary pre-ordering, and delivery management, while also integrating a gamified milestone tracker that generated a downloadable and shareable "Kapampangan Heritage Champion" digital certificate of achievement upon the completion of qualifying culinary journeys.

Design Phase. The Design Phase translated the approved system requirements into detailed architectural and technical designs by developing the system's structural framework, database architecture, and user interface prototypes. During this phase, the proponents created system flowcharts, relational PostgreSQL database schemas managed through Django's Object-Relational Mapping (ORM), data flow diagrams, and interactive web interface wireframes using Figma. The database structure was designed to organize and partition master records across all municipalities and component cities of Pampanga, ensuring efficient data storage, fast query processing, and scalable system performance. Furthermore, the user interface wireframes were designed to incorporate the Live Navigation Viewport, which featured real-time path tracing, dynamic estimated time of arrival (ETA) updates, and highlighted historical, cultural, and ecological attractions situated along the active culinary route. The design also included the Merchant Booking and Logistics Dashboard, which allowed partner establishments to manage incoming reservations, monitor pre-ordered menus, and coordinate delivery address dispatches. In addition, the interface featured the "Kapampangan Heritage Champion" Achievement Hub, where users were able to view their milestone accomplishments and access a downloadable and printable digital certificate upon completing qualifying culinary journeys. The responsive user interface was designed to provide intuitive navigation and a seamless user experience across desktop and mobile web browsers while minimizing cognitive load for travelers exploring Pampanga's culinary destinations.

Develop Phase. The Develop Phase involved the actual coding, integration, and implementation of the proposed system based on the approved design specifications. During this phase, the proponents developed the Kanyamanan platform using React.js for the presentation layer and the Django (Python) Web Framework for the backend API and database orchestration layer. The backend development leveraged the Django REST Framework (DRF) to design secure and scalable application programming interfaces (APIs) for route

optimization, user authentication, menu nutritional processing, and communication between the frontend and backend components. In addition, the development process integrated Python-based spatial routing libraries and Google Maps API wrappers to compute live GPS positions, generate optimized route sequences, and provide real-time estimated time of arrival (ETA) updates for culinary travelers. The system also utilized Django state management and signals to dynamically coordinate table reservations, monitor real-time menu pre-orders, and manage delivery configurations within the centralized database. Furthermore, a PDF generation module, such as ReportLab or WeasyPrint, was integrated into the Django framework to automatically generate high-resolution, verifiable "Kapampangan Heritage Champion" digital certificates once users successfully completed the required culinary milestones. These integrated technologies collectively supported the development of a secure, scalable, and user-centered web platform capable of delivering intelligent culinary itinerary planning, merchant management, and health informatics services for culinary tourists in Pampanga.

Test Phase. The Test Phase focused on identifying, documenting, and correcting errors to verify that the system satisfied all functional requirements. The proponents conducted unit testing (validating isolated backend routes), integration testing (verifying database transaction flow), and performance testing (profiling the spatial routing engine to maintain computational latencies under two seconds). User Acceptance Testing (UAT) was also conducted with representative test groups to evaluate the functional performance of the software. Technical issues, routing exceptions, and prediction margins identified during testing were documented and resolved before the system was deployed.

Deploy Phase. The Deploy Phase involved publishing the validated codebase to a staging or production web server for actual use. During this phase, the Kanyamanan web application was deployed to a secure, centralized cloud

hosting environment (such as AWS or Vercel/Heroku) configured with HTTPS/SSL communication layers.

The deployment configuration leveraged Django static and media file hosting structures (coupled with cloud storage services like AWS S3) to efficiently store and deliver user-uploaded culinary images, restaurant menus, and programmatically generated PDF achievement certificates. Progressive Web App (PWA) client-side caching protocols ensured that map routing details, cached itineraries, and landmark coordinates remained visible in offline or low-signal zones.

Review Phase. The Review Phase closed the cycle by gathering feedback from the Pampanga Provincial Tourism Officer, verified merchants, and tourists to plan subsequent sprints. After deployment, stakeholders were asked to assess the system in terms of usability, efficiency, and functionality. Their feedback was documented and analyzed during sprint retrospective meetings to refine database indexes, improve algorithm parameter boundaries, and adjust the user interface. This ensured that the system remained relevant and adaptable to the evolving needs of the provincial tourism sector.

Launch (The Output). The Launch Phase represented the successful output of the sprint cycle, where the fully functional, production-ready software increment was deployed for immediate stakeholder utilization (Project Management Institute, 2025). During this phase, the Kanyamanan system was made accessible to the public and local culinary MSMEs, providing verified merchants with access to claimable profile settings while allowing tourists to officially generate, save, and utilize optimized culinary-cultural routes across all municipalities and component cities of the Province of Pampanga.

In this final stage, verified restaurant merchants were able to officially manage local reservations, pre-orders, and delivery options, while consumers

used live GPS tracking to reach their destinations and printed their verified "Kapampangan Heritage Champion" achievements.

3.3 Data Gathering Procedures

This section described the step-by-step procedures that were followed in gathering the data needed for the development and evaluation of the proposed system. It consisted of three phases: preparation, implementation, and post-collection. Data gathering methods included surveys, questionnaires, and document analysis. Ethical considerations such as informed consent, confidentiality, and approval from relevant authorities were also observed. Furthermore, this section explained how the system requirements were identified and how the collected data were transformed into system design inputs. The data gathering process was conducted in three phases: preparation, implementation, and post-collection.

In the preparation phase, the proponents secured institutional clearance by submitting official endorsement letters to the University of the Assumption Research Ethics Board (UAREB) to ensure full ethical compliance. Structured questionnaires based on the ISO/IEC 25010 standard were developed. These instruments were validated through Content Validity Index (CVI) assessments by a panel of three Information Technology experts to verify their alignment and reliability. Informed consent forms were prepared to clearly outline the data privacy protections established under the Data Privacy Act of 2012, commonly known as Republic Act No. 10173 (Cado, 2024).

During the implementation phase, document analysis was performed on provincial tourism registries, regional maps, and Kapampangan restaurant menus to collect baseline geolocations, operating hours, and dish ingredients. Simultaneously, structured surveys were distributed to culinary tourists traveling along major transit corridors to gather information regarding user preferences and travel-related challenges.

In the post-collection phase, quantitative data obtained from the tourist surveys were encoded, cleaned, and analyzed using appropriate descriptive statistical tools. The pre-seeded multi-vendor catalog—composed of verified coordinates, price points, operating hours, and dish nutritional compositions (cross-referenced with the DOST-FNRI database)—was finalized and securely migrated into the MongoDB database layer, thereby creating the foundational dataset for the proposed decision support platform.

The data gathered from surveys and document analysis were consolidated and verified before being utilized in the development of the proposed system. The validated information served as the basis for identifying the functional requirements, designing the database, developing the system features, and evaluating the overall quality and acceptability of the Kanyamanan platform.

3.4 Sources of Data

The study utilized both primary and secondary sources of data to ensure the accuracy, reliability, and completeness of the information gathered for the development and evaluation of the proposed Kanyamanan platform.

Primary data were obtained directly from the respondents through survey questionnaires and consultations conducted during the requirements-gathering and system evaluation phases. The respondents consisted of local culinary micro, small, and medium enterprise (MSME) operators, culinary tourists and same-day excursionists, representatives from the Provincial Tourism Office, and Information Technology specialists. These stakeholders provided firsthand information regarding user requirements, existing challenges, operational practices, preferred system features, and evaluations of the proposed platform in terms of functionality, usability, efficiency, and overall acceptability.

Secondary data were collected from credible and authoritative sources to supplement and validate the primary data. These included published journals, books, conference proceedings, government reports, official tourism publications, food composition tables, online databases, and related studies on culinary tourism, health informatics, route optimization, recommender systems, and web-based information systems. Additional information was obtained from the official websites of relevant government agencies, including the Provincial Tourism Office, to support the development of the culinary destination database, nutritional information, and other system components.

The integration of primary and secondary data sources enabled the proponents to obtain both empirical insights from stakeholders and verified information from reliable references. These data collectively served as the foundation for the design, development, implementation, and evaluation of the Kanyamanan platform, ensuring that the proposed system was relevant, evidence-based, and aligned with the objectives of the study.

3.5 Sampling Technique

The research applied a mixed sampling methodology that integrated statistical calculations for identifying end-users with geographical stratified purposive sampling for stakeholder and expert evaluations.

The Raosoft sample size calculator was employed to determine the appropriate number of end-users. Assuming an extensive baseline population of culinary tourists and local excursionists, the calculation utilized a 90% confidence level, a 10% margin of error, and a 50% response distribution. This yielded a minimum statistical requirement of 68 participants. To guarantee a robust dataset, the researchers set a final target of 70 culinary tourists and local excursionists. Since the data collection process relied entirely on online surveys, a targeted digital screening method was implemented. Preliminary questions were used to filter the respondents to verify that they actively engaged in food trips or dining excursions within the province. This approach ensured that the sample captured a comprehensive user base, encompassing both inbound tourists visiting from outside Pampanga and local residents who regularly explored the regional culinary landscape. Both groups actively experienced the challenges of discovering dining venues and navigating multi-stop food trips.

To mitigate localized data bias and verify that the multi-stop routing engine, live GPS tracker, and reservation functionalities of the proposed platform functioned efficiently throughout the province, a Geographical Quota Distribution was integrated into the online tourist sampling process. The 70 culinary tourist and local excursionist respondents were categorized according to their travel experiences across the four legislative districts of the Province of Pampanga. Implementing this district-based quota distribution ensured that the dynamic route configurations, attraction mapping, live GPS navigation, and transactional functionalities of the proposed Kanyamanan platform were comprehensively evaluated across all geographic sectors of Pampanga, thereby improving the representativeness and reliability of the study's findings.

For the stakeholder and expert evaluations, the research relied strictly on purposive sampling. This non-probability technique allowed for the intentional selection of respondents based on their direct involvement, experience, and knowledge regarding the research objectives. The evaluation panel consisted of 10 Local Culinary MSME Operators chosen from various municipalities to assess the claimable merchant portal, validate the pre-seeded operational data, and offer insights into business workflows. Furthermore, 3 Provincial Tourism Office Representatives were selected to evaluate how well the system aligned with administrative goals, destination promotion, and regional tourism initiatives. Finally, 5 Information Technology Specialists conducted the technical evaluation of the software using the ISO/IEC 25010 Software Product Quality Model. Their assessment covered functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability.

In total, the study engaged 88 respondents to guarantee a technically rigorous, spatially representative, and comprehensive evaluation of the Kanyamanan system. The collective feedback from these participants established the empirical foundation needed to validate the overall acceptability, usability, and functionality of the proposed web platform.

3.6 Interpretation of Data

This section explained how the collected data were analyzed and interpreted. It also described the statistical tools and evaluation methods that were used to analyze the gathered data and assess the proposed system. For system evaluation, descriptive statistical tools such as the Weighted Mean and Standard Deviation were used to analyze the responses gathered through the four-point Likert scale, while the ISO/IEC 25010 Software Product Quality Model served as the framework for evaluating the quality characteristics of the proposed system. The interpretation of the collected data aligned with the objectives of the study and the evaluation criteria established for the proposed system.

The quantitative evaluation data gathered from the Information Technology specialists, local culinary MSME operators, culinary tourists and same-day excursionists, and representatives from the Provincial Tourism Office were analyzed using descriptive statistical tools. The overall quality, performance, and user acceptability of the Kanyamanan platform were evaluated using the Weighted Mean and Standard Deviation.

The Weighted Mean was used to determine the average evaluation of the respondents for each of the ISO/IEC 25010 software quality characteristics. This statistical tool provided the basis for determining the level of agreement regarding the functionality, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability of the proposed system.

The Standard Deviation was used to determine the consistency or variability of the respondents' evaluations. A lower standard deviation indicated that the respondents' ratings were more consistent, while a higher standard deviation indicated greater variation in their responses.

The evaluation was conducted using a four-point Likert scale, which encouraged respondents to provide a definite assessment by eliminating a neutral response. The computed weighted mean values were interpreted using the statistical ranges and verbal interpretations presented in Table 3.1.

Table 3.1: 4-Point Likert Scale and Verbal Interpretation Matrix

Scale Value	Statistical Range	Verbal Response	Verbal Interpretation
4	3.26 – 4.00	Strongly Agree	Very High
3	2.51 – 3.25	Agree	High
2	1.76 – 2.50	Disagree	Low
1	1.00 – 1.75	Strongly Disagree	Very Low

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